

Processor Basics

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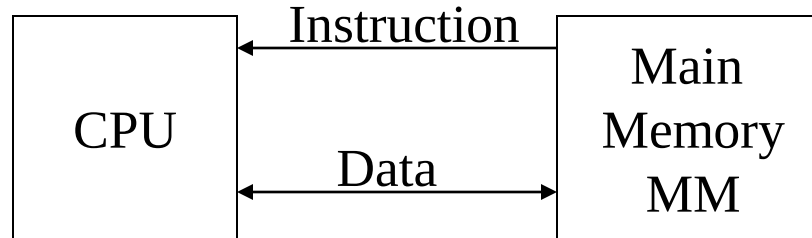
CPU Organization

The primary function of the CPU is to execute sequences of instructions, which are stored in memory.

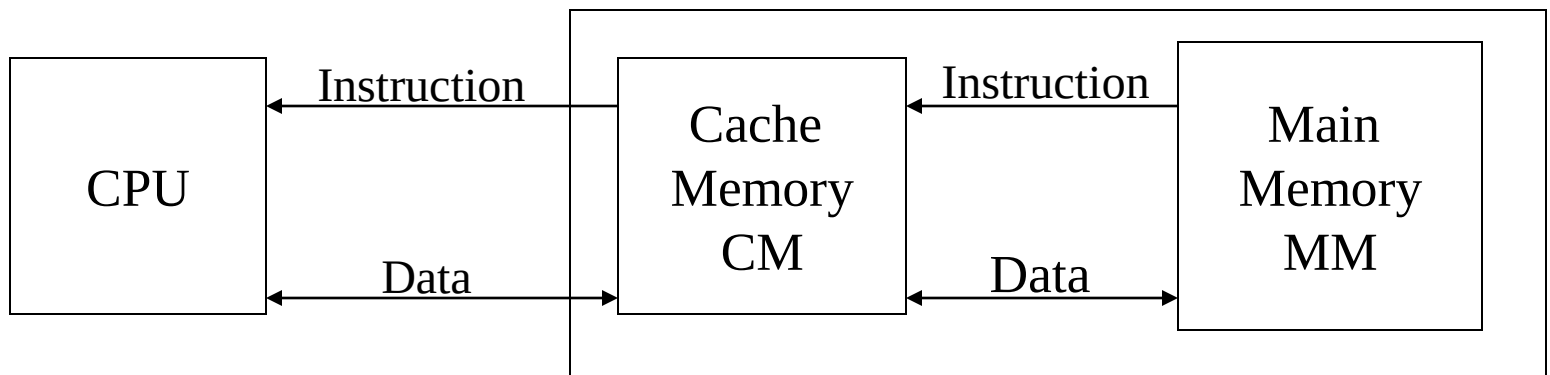
The execution is carried out in three steps:

- The CPU transfer instruction and when necessary, their input data (operand) from main memory to registers in the CPU.
- The CPU executes the instruction in their stored sequence except when the execution sequence is explicitly altered by a branch instruction.
- When necessary, the CPU transfers data (results) from the CPU registers to main memory.

External Communication

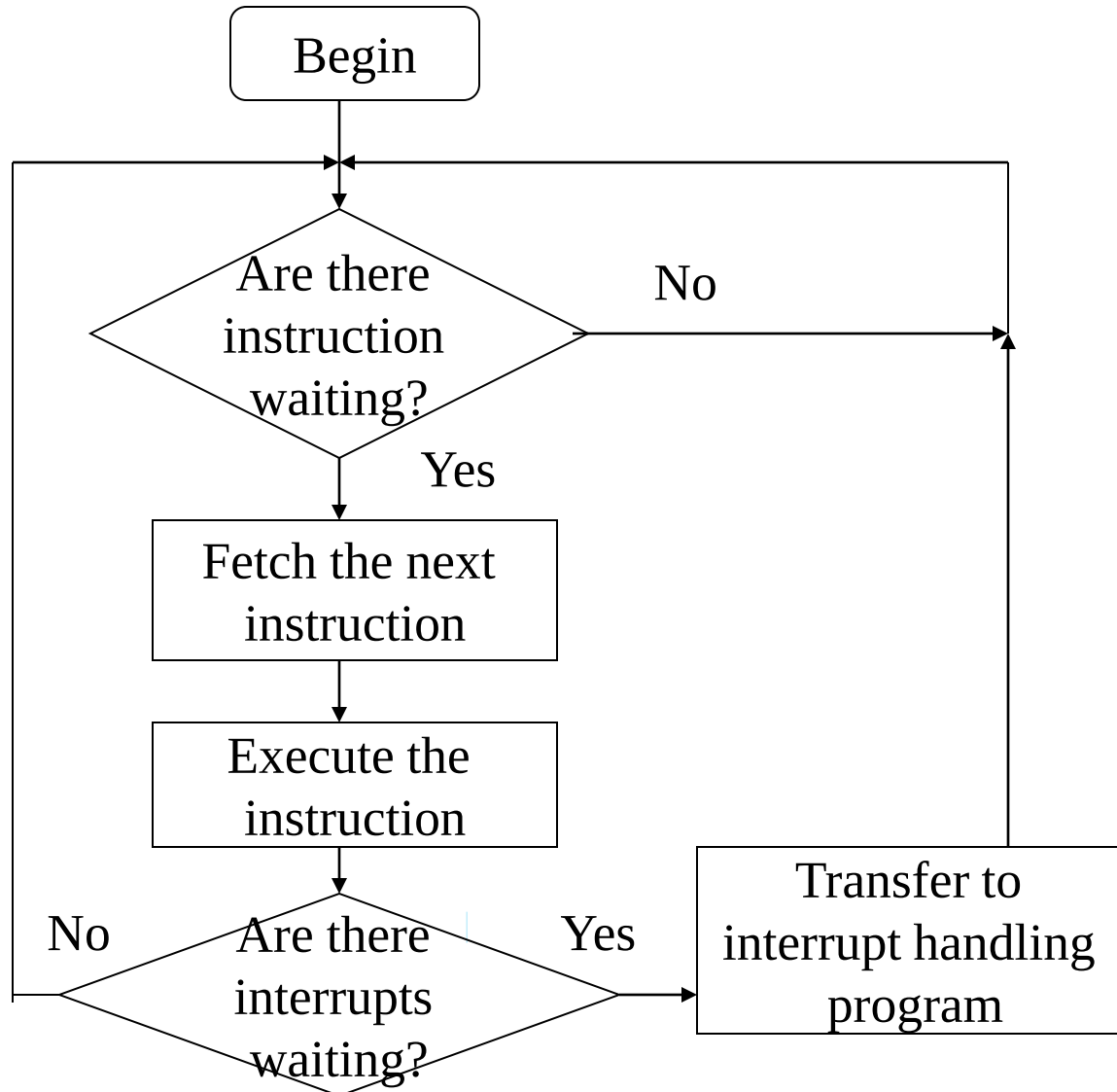


Processor-memory communication without a cache



Processor-memory communication with a cache

CPU Operation



CT CPU Operation

- **Fetch**: The processor usually gets instructions from RAM.
- **Decode**: A decoder translates an instruction into signals that the computer's other parts can use.
- **Execute**: Each component receives the now-decoded instructions.
- **Store**: After the execute step the instructions are ready to store in the memory.

Types of Processors

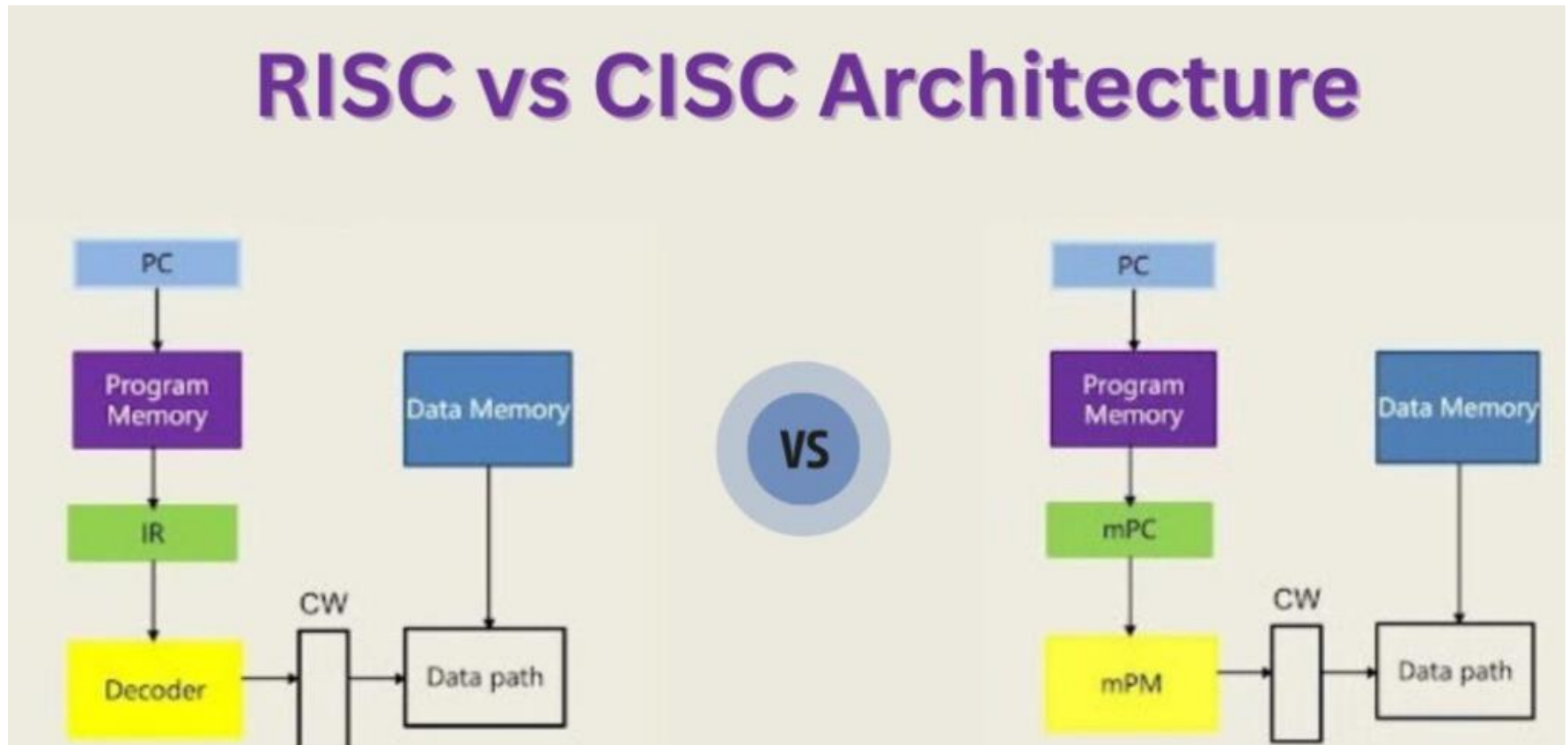
- **Single Core Processors:** The oldest type of computer CPUs is single core CPU. These CPUs were used in the 1970s. these CPUs only have a single core that perform different operations. This means that the single core CPU can only process one operation at a single time. single core CPU is not suitable for multitasking.
- **Dual-Core Processors:** Dual-Core CPUs contain a single Integrated Circuit with two cores. Each core has its cache and controller. These controllers and cache work as a single unit. dual core CPUs can work faster than the single-core processors.
- **Quad-Core Processors:** Quad-Core CPUs contain two dual-core processors present within a single integrated circuit (IC) or chip. A quad-core processor contains a chip with four independent cores. These cores read and execute various instructions provided by the CPU. Quad Core CPU increases the overall speed for programs. Without even boosting the overall clock speed it results in higher performance.

Which Processor Should You Select?

- When upgrading processor, check if motherboard supports the new one. Intel motherboards don't work with AMD processors and vice versa. Some motherboards only support old processors, so new one may not be compatible. Certain motherboards only support latest processors. Make sure motherboard and new processor match. If not, either get new motherboard for that processor or new processor for existing motherboard. If compatible, here's a basic performance breakdown from AMD and Intel based on your needs:
- CPUs with AMD Ryzen 3/Intel Core i3 architecture are typically found in the most affordable models.
- This is much more of a "everyday" type of CPU, AMD Ryzen 5/Intel Core i5. It can perform really well and reasonable.
- These chips—AMD Ryzen 7 and Intel Core i7—are perfect for PC gaming and light-duty creative content creation.
- Regardless of price, the enthusiast market expects the finest performance, and AMD Ryzen 9/Intel Core i9 is the best option currently available.

RISC vs CISC Architecture

There are two approaches to processor architecture are represented by the terms “Reduced Instruction Set Computing (**RISC**)” and “Complex Instruction Set Computing (**CISC**)” respectively.



RISC vs CISC Architecture

RISC and CISC are two different types of computer architectures that are used to design the microprocessors that are found in computers. The fundamental difference between RISC and CISC is that **RISC (Reduced Instruction Set Computer)** includes simple instructions and takes one cycle, while the **CISC (Complex Instruction Set Computer)** includes complex instructions and takes multiple cycles.

What is RISC?

In the RISC Architecture, the instruction set of the computer system is simplified to reduce the execution time. RISC architecture has a small set of instructions that generally includes register-to-register operations.

The RISC architecture uses comparatively a simple instruction format that is easy to decode. The instruction length can be fixed and aligned to word boundaries. RISC processors can execute only one instruction per clock cycle.

The following are some important **characteristics** of a RISC Processor:

- A RISC processor has a few instructions.
- RISC processor has a few addressing modes.
- In the RISC processor, all operations are performed within the registers of the CPU.
- RISC processor can be of fixed-length.
- RISC can be hardwired rather than micro-programmed control.
- RISC is used for single-cycle instruction execution.
- RISC processor has easily decodable instruction format.

What is CISC?

A CISC processor has a variable-length instruction format. In this processor architecture, the instructions that require register operands can take only two bytes.

In a CISC processor, the execution of instructions may take a varying number of clock cycles. The CISC processor also provides direct manipulation of operands that are stored in the memory.

The important **characteristics** of a CISC processor architecture:

- CISC can have variable-length instruction formats.
- It supports a set of a large number of instructions, typically from 100 to 250 instructions.
- It has a large variety of addressing modes, typically from 5 to 20 different modes.
- CISC has some instructions which perform specialized tasks and are used infrequently.

RISC vs CISC?

RISC	CISC
Small and simple instruction set	Large and complex instruction set
Fixed-length instructions	Variable-length instructions
Usually 1 cycle per instruction	Multiple cycles per instruction
More general-purpose registers	Fewer registers
Simple and limited addressing modes	Complex and many addressing modes
Larger code size (more instructions)	Smaller code size (fewer instructions)
Simple hardware, easier pipelining	Complex hardware, harder pipelining
Examples: ARM, RISC-V, MIPS	Examples: x86, Intel 80386

H.W.

CT Which type of instruction set architecture is used in modern processors such as Intel Core i7 — RISC, CISC, or a combination of both? Explain how it works.

Short Ans: (Try to learn in details)

Modern Intel Core i7 processors are based on **CISC (Complex Instruction Set Computer)** architecture — specifically, the **x86/x86-64** family. However, **internally**, they use **RISC-like micro-operations (μops)** for execution.

Here's how it works:

The processor **receives complex CISC instructions** (like “move data,” “multiply,” etc.).

Inside the CPU, these are **translated (decoded)** into **simpler RISC-style micro-operations**.

These micro-operations are then executed by a **RISC-like core** for speed and efficiency.

Ref.

1. <https://www.tutorialspoint.com/difference-between-risc-and-cisc>
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3. <https://www.geeksforgeeks.org/computer-organization-architecture/computer-organization-risc-and-cisc/>
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